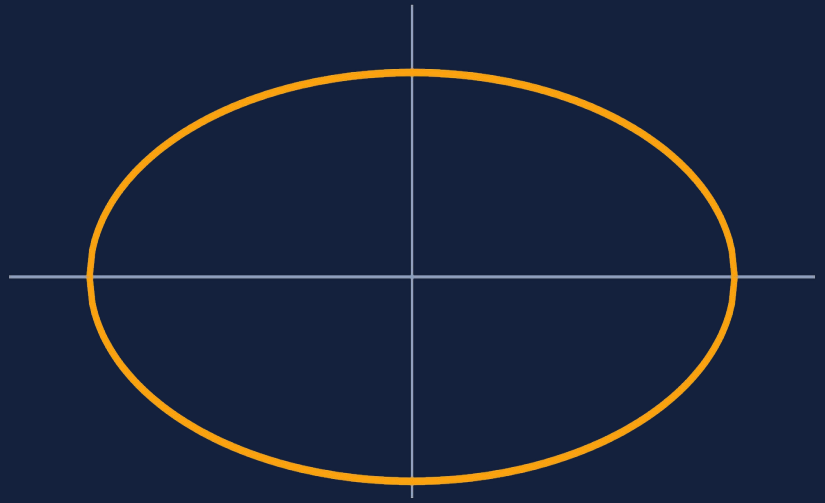
Stability and Routh–Hurwitz Criterion



Control Systems (22EEG53)

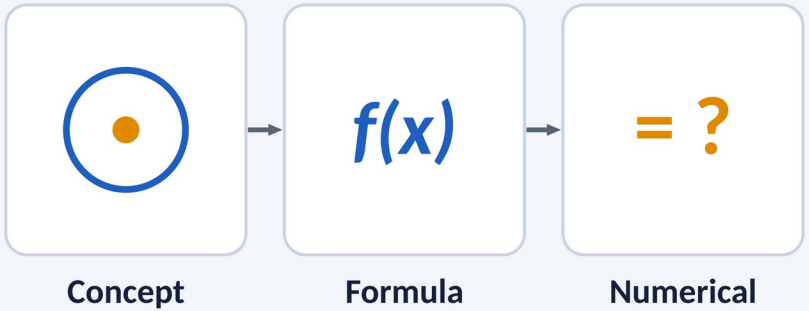
Unit 4: Stability of Systems

Topic: Characteristic equation, root locations, absolute and relative stability, Routh–Hurwitz criterion

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Learning Outcomes

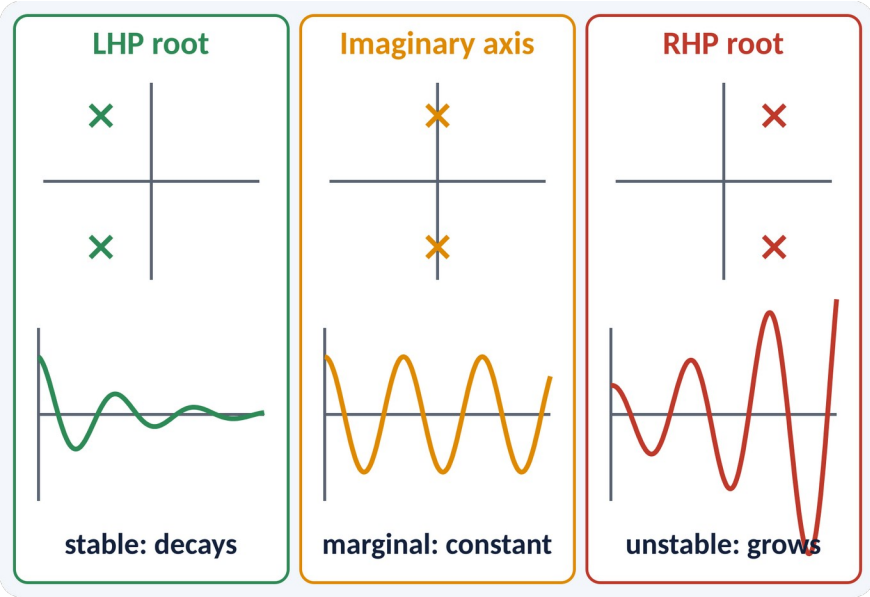
- Relate root locations to stability.
- Distinguish absolute and relative stability.
- Build the Routh array and read stability from it.
- Find the range of K for a stable loop.



Linked to CO4
Understand stability using Routh–Hurwitz and root locus.

Recap: What You Already Know

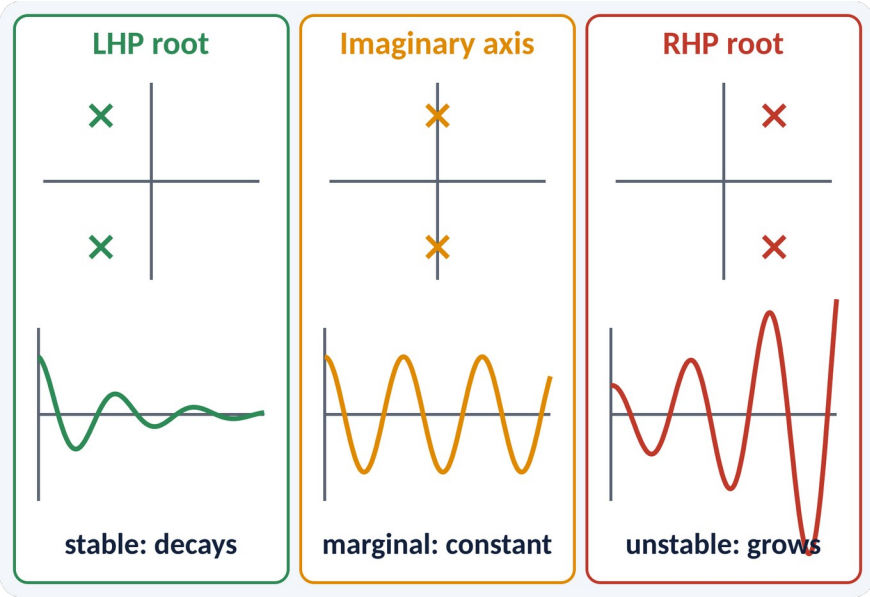
- Poles are roots of the denominator.
- Pole location fixes the time response.
- Real part σ gives decay or growth e^(σt).
- Closed loop: T(s) = G/(1 + GH).



Closed-loop poles: roots of 1 + G(s)H(s) = 0 (characteristic equation)

Motivation: Why Test Stability?

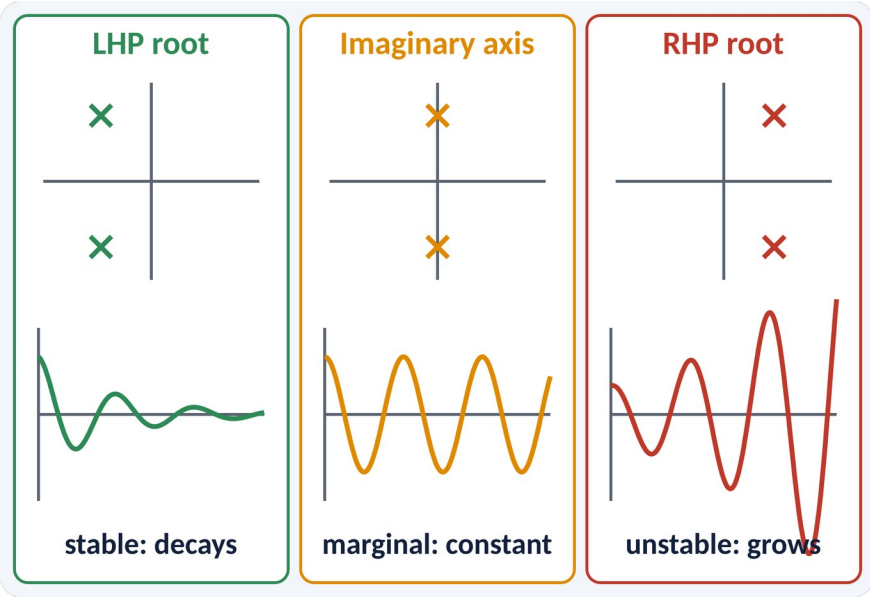
- Unstable aircraft or robot: output grows without bound.
- Feedback can turn a stable plant unstable.
- High gain often causes oscillation.
- Solving high-order polynomials is hard.



Routh's test tells stability without computing the roots

Characteristic Equation and Root Location

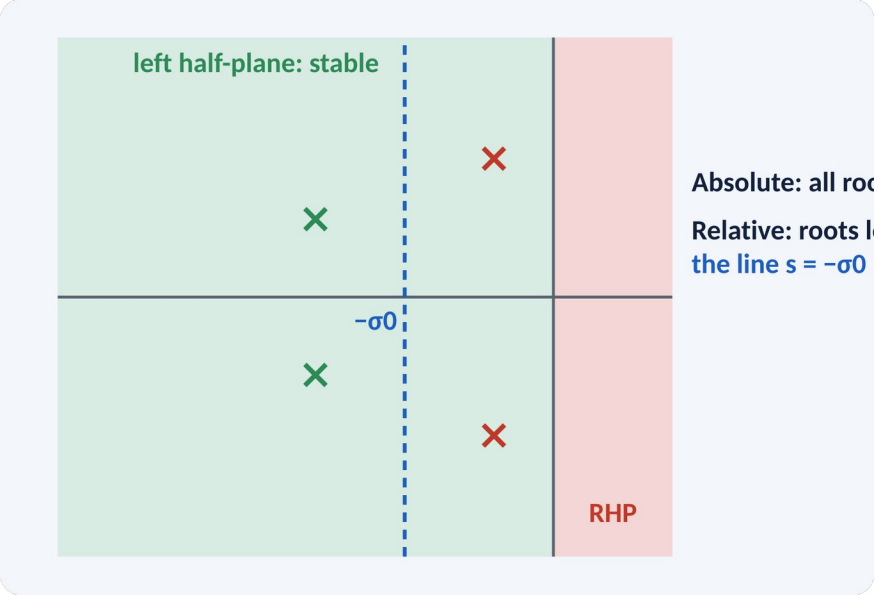
- Stable: all roots in the left half-plane.
- Marginally stable: simple roots on the jω axis.
- Unstable: any root in the right half-plane.
- Repeated jω roots also give instability.



Characteristic equation: $a_0 s^n + a_1 s^{n-1} + \ldots + a_n = 0$
BIBO stable ⇔ all roots have negative real part

Absolute and Relative Stability

- Absolute: are all roots in the LHP? yes or no.
- Relative: how far are the roots from the jω axis?
- Roots near the axis give poor damping.
- Shift the axis by σ0 to test relative stability.



Substitute s = z − σ0 and apply Routh to the z polynomial

Necessary Condition for Stability

- All coefficients must be present.
- All coefficients must have the same sign.
- If violated, the system is not stable.
- If satisfied, it is still not enough.

$s^3 + 2s^2 + 3s + 4$
all coefficients positive: go on to test

$s^3 - 2s^2 + 3s + 4$
negative coefficient: unstable

$s^3 + 2s^2 + 4$
missing s term: not stable

$s^3 + s^2 + 2s + 8$
all positive but still unstable!

Necessary, not sufficient: $s^3 + s^2 + 2s + 8$ has positive coefficients but is unstable

Routh Array Construction

- Write coefficients in two rows: s^4, s^3.
- Each next row from the two rows above.
- Cross-multiply, subtract, divide by the pivot.
- Continue down to the s^0 row.
- First column carries the information.

$$a0\,s^4 + a1\,s^3 + a2\,s^2 + a3\,s + a4 = 0$$

s^4	a0	a2	a4
s^3	a1	a3	0
s^2	b1	b2	0
s^1	c1	0	
s^0	d1		

$$b1 = (a1a2 - a0a3)/a1 \quad b2 = (a1a4 - a0 \cdot 0)/a1$$
$$c1 = (b1a3 - a1b2)/b1 \quad d1 = a4$$

$b1 = (a1a2 - a0a3)/a1$; $c1 = (b1a3 - a1b2)/b1$
Number of rows = degree n + 1

Reading the Routh Table

- Look only at the first column.
- All same sign: all roots in LHP.
- Each sign change: one root in RHP.
- Number of sign changes counts RHP roots.

s^4	1	3	5
s^3	2	4	
s^2	1	5	
s^1	−6		
s^0	5		

2 sign changes (+ → − → +) = 2 roots in RHP

Routh–Hurwitz: stable ⇔ first column has no sign change
Example $s^4 + 2s^3 + 3s^2 + 4s + 5$: column 1, 2, 1, −6, 5 → 2 changes

Special Case 1: Zero in First Column

- A zero as pivot stops the calculation.
- Replace 0 by a small positive ε.
- Finish the table in terms of ε.
- Let ε → 0 and count sign changes.
- Alternative: substitute s = 1/z.

Case 1: A zero in the first column

Replace 0 by ε (small, positive); complete the table; let ε → 0 and count sign changes.

Example: $s^3 - 3s + 2 = 0$

s^3: 1, −3; s^2: 0→ε, 2; s^1: (−3ε − 2)/ε → −∞;
s^0: 2 → 2 sign changes.

Example: $s^3 - 3s + 2$: s^2 row (ε, 2) → $s^1 = -\infty$ → 2 sign changes

Special Case 2: Row of Zeros

- A whole row becomes zero.
- Form auxiliary polynomial A(s) from the row above.
- Differentiate A(s); use its coefficients as the new row.
- Roots of A(s) are roots of the system too.
- A pair on jω: marginally stable.

Case 2: An entire row of zeros

Form the auxiliary polynomial A(s) from the row above; differentiate it and replace the zero row.

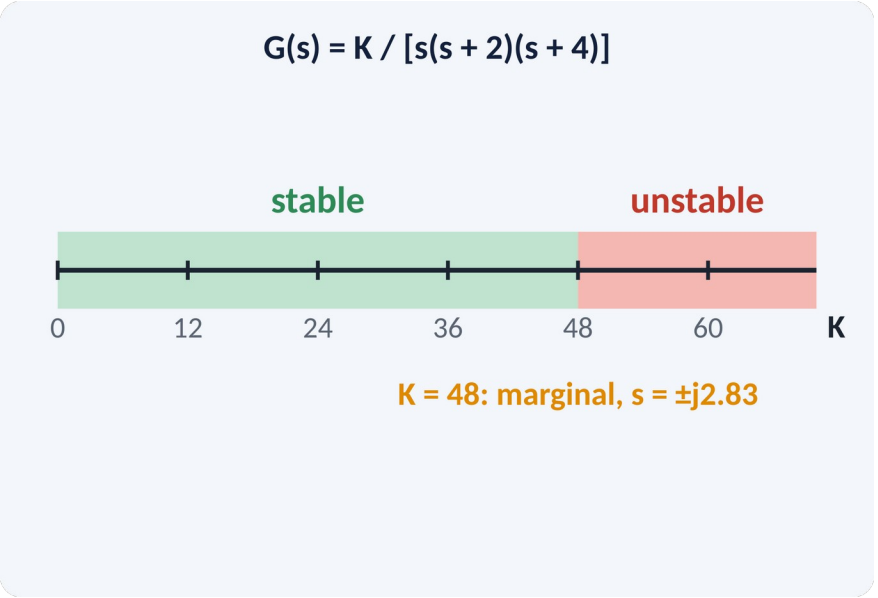
Meaning: Roots of A(s) are symmetric

Pairs ±σ or ±jω lie on the roots of A(s); ±jω pair means marginal stability.

Row of zeros ⇒ roots symmetric about the origin (±σ or ±jω)

Range of Gain for Stability

- Put K in the characteristic equation.
- Build the array with K as a symbol.
- Every first-column term must be positive.
- Solve the inequalities for K.
- Boundary values give marginal stability.



Example: $s^3 + 6s^2 + 8s + K = 0 \rightarrow 0 < K < 48$
At K = 48 the roots are $s = \pm j\sqrt{8} = \pm j2.83$

Solved Example 1: Routh Array

- Test $s^4 + 2s^3 + 3s^2 + 4s + 5 = 0$ for stability.
- 1. All coefficients positive: continue.
- 2. s^4: 1, 3, 5; s^3: 2, 4.
- 3. s^2: (2×3 − 1×4)/2 = 1; (2×5 − 0)/2 = 5.
- 4. s^1: (1×4 − 2×5)/1 = −6.
- 5. s^0: 5.
- 6. Column: 1, 2, 1, −6, 5 → two sign changes.



Answer: unstable; two roots in the RHP (≈ 0.29 ± j1.42)

Solved Example 2: Range of K

- Unity feedback, G = K/[s(s + 2)(s + 4)]. Find K for stability and the frequency at marginal stability.
- 1. Characteristic eq.: $s^3 + 6s^2 + 8s + K = 0$.
- 2. Array: s^3: 1, 8; s^2: 6, K; s^1: (48 − K)/6; s^0: K.
- 3. $s^1 > 0 \rightarrow K < 48$; $s^0 > 0 \rightarrow K > 0$.
- 4. At K = 48: $6s^2 + 48 = 0 \rightarrow s = \pm j2.83$ rad/s.

s^3	1	8
s^2	6	K
s^1	(48 − K)/6	0
s^0	K	

$s^1 > 0 \Rightarrow K < 48$; $s^0 > 0 \Rightarrow K > 0$

$0 < K < 48$

Answer: stable for 0 < K < 48; sustained oscillation at 2.83 rad/s when K = 48

Common Mistakes and Exam Tips

Common mistakes

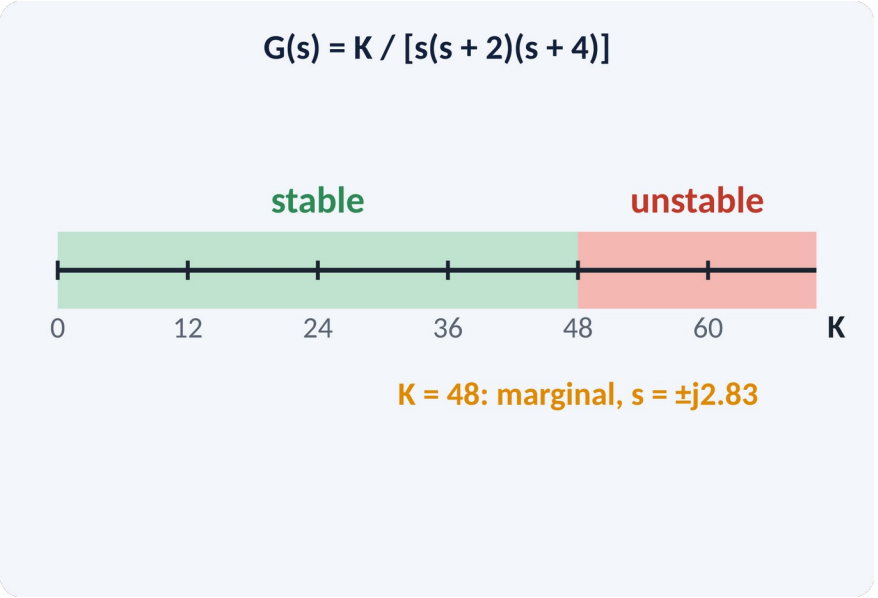
- Concluding stability from positive coefficients alone.
- Arithmetic error in the cross products.
- Counting sign changes in the whole row instead of the first column.
- Ignoring special cases (zero pivot, zero row).
- Forgetting to include K in the s^0 term.

Exam tips

- Check necessary condition first.
- Divide each row by a positive number to simplify.
- Count first-column sign changes only.
- Write the stable range as an inequality in K.

Summary

- Stable ⟺ all roots of 1 + GH = 0 in the LHP.
- Necessary: all coefficients present and same sign.
- Routh array: sign changes in column 1 = RHP roots.
- Zero pivot → ε; zero row → auxiliary polynomial.
- Use Routh to find the K range for stability.



Stability range of K from the first column: all entries > 0

Quick Quiz

Q1. A system with a root in the RHP is:
(a) stable (b) marginally stable (c) unstable (d) critically damped

Ans: (c)

Q2. $s^3 + 2s^2 + 3s + 4 = 0$ first column of Routh array has:
(a) 1, 2, 1, 4 (b) 1, 2, −1, 4 (c) 1, 2, 0, 4 (d) 1, −2, 1, 4

Ans: (a)

Q3. Sign changes in the first column give:
(a) poles on jω (b) RHP roots (c) LHP roots (d) zeros

Ans: (b)

Q4. A row of zeros means:
(a) always unstable (b) symmetric roots, use auxiliary polynomial (c) always stable (d) type 2

Ans: (b)

Q5. $s^3 + 6s^2 + 8s + K$: stable for:
(a) $K < 6$ (b) $0 < K < 48$ (c) $K > 48$ (d) any K

Ans: (b)

References

1. **Benjamin C. Kuo, Automatic Control Systems**
   7th Ed., PHI, 2007 (Units 3, 4, 5 textbook)

2. **M. Gopal, Control Systems Principles and Design**
   4th Ed., TMH, 1999 (reference, Units 4, 5)

3. **P. S. Satyanarayana, Concepts of Control Systems**
   2nd Ed., Dynaram Publishers, Bangalore, 2001 (reference)

Next lecture: root locus concept and construction rules